Does an inverse exist?

1

a No

b Yes

c Yes

d Yes

2

a Yes

b No

c Yes

d Yes

e No

3

a Yes

b No

c Yes

d Yes

e Yes

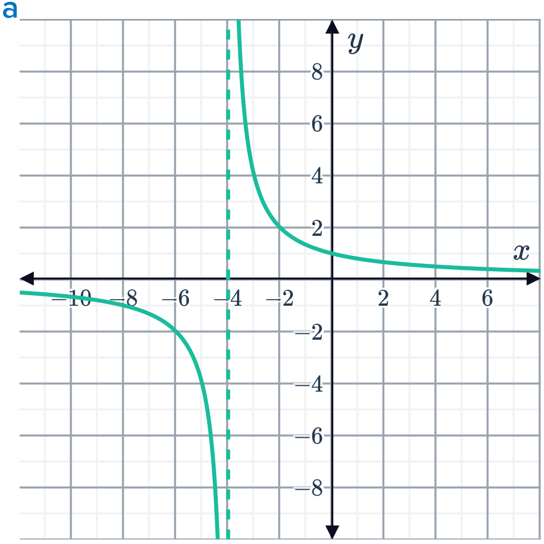
f Yes

g Yes

h No

4

a



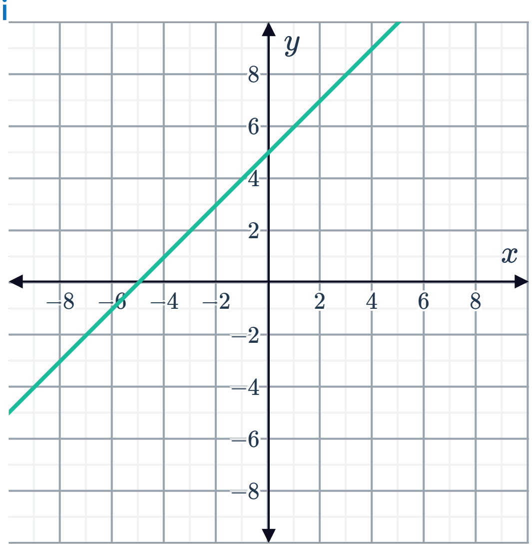
b Yes

c $y = \dfrac{4}{x} - 4$

5

a

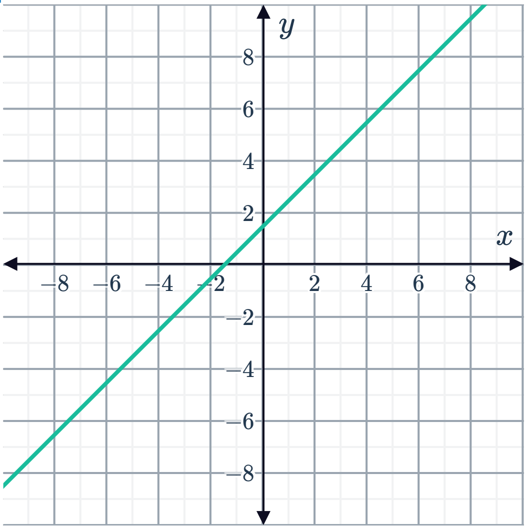
i



ii Yes

iii Yes

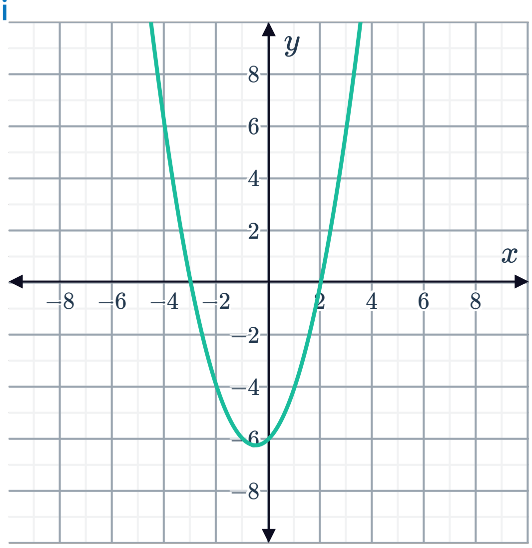
b

i


ii Yes

iii Yes

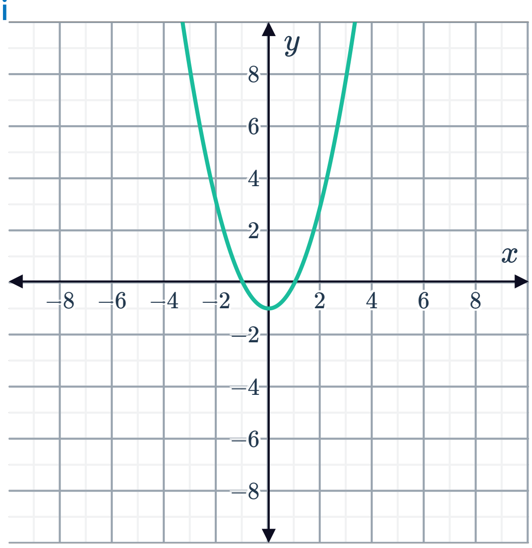
c

i


ii No

iii No

d

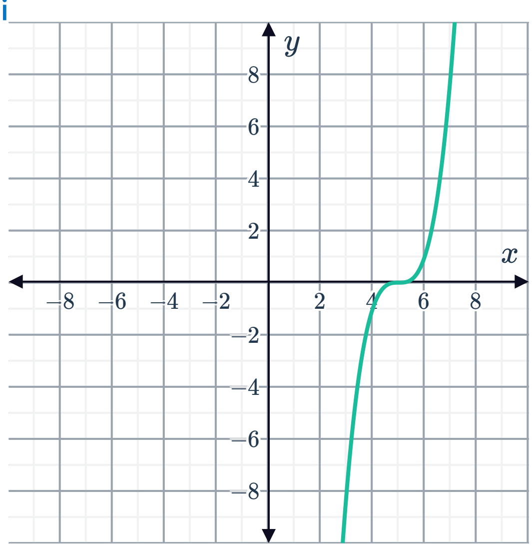
i


ii No

iii No

e

i

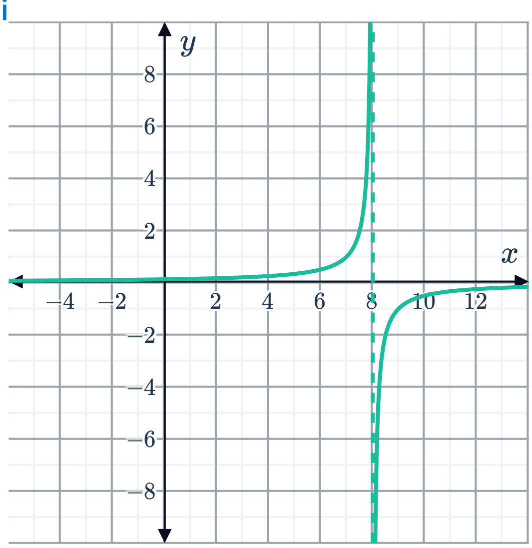


ii Yes

iii Yes

f

i



ii Yes

iii Yes

Restricting the domain

6

a

$$f(x) = \begin{cases} (x-2)^2 - 6, & x < 2 \\ (x-2)^2 - 6, & x \geq 2 \end{cases}$$

b

$$f(x) = \begin{cases} g(x), & x \leq -1 \\ g(x), & -1 < x < 6 \\ g(x), & x \geq 6 \end{cases}$$

7 $(-\infty, 7]$

8

a $y = \sqrt{x+6} + 4$

b Domain: $[-6, \infty)$, Range: $[4, \infty)$

9

a $[-2, \infty)$

b $y = \sqrt{x+9} - 2$

c Domain: $[-9, \infty)$, Range: $[-2, \infty)$

10

a $[-4, \infty)$

b $y = \sqrt[4]{x} - 4$

c Domain: $[0, \infty)$, Range: $[-4, \infty)$

11

a $(-\infty, 5]$

b $y = -\sqrt{-x} + 5$

c Domain: $(-\infty, 0]$, Range: $(-\infty, 5]$

12

a $f(x) = (x+5)^2 - 2$

b $[-5, \infty)$

c $y = \sqrt{x+2} - 5$

d Domain: $[-2, \infty)$, Range: $[-5, \infty)$

13

a $f(x) = (x+1)(x+3)$

b $x = -2$

c $[-2, \infty)$

d $y = \sqrt{x+1} - 2$

e Domain: $[-1, \infty)$, Range: $[-2, \infty)$

14

a Domain: $(-\infty, 8]$, Range: $[0, \infty)$

b $y = -x^2 + 8$

c $[0, \infty)$

d The domain of $f(x)$ is equivalent to the range of $f^{-1}(x)$ while the range of $f(x)$ is equivalent to the domain of $f^{-1}(x)$.

15

a Domain: $[5, \infty)$, Range: $[8, \infty)$

b Yes

c $y = (x-8)^2 + 5$

d $[8, \infty)$

e The domain of $f(x)$ is equivalent to the range of $f^{-1}(x)$.

16

a Domain: $[a, \infty)$, Range: $[b, \infty)$

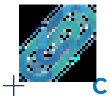
b $y = (x-b)^{\frac{1}{4}} + a$

c Domain: $[b, \infty)$, Range: $[a, \infty)$

17

a $[a, \infty)$

b $y = \sqrt{-x-a} + a$



c $a \leq \frac{1}{8}$

18

a $b = -6$

b $c = 7$

Applications

19

a No

b Yes

c No

d No

20

a 7.8 tonnes per capita

b $t = \left(\frac{M - 7.8}{2.95}\right)^2$

c 7.0 years

21

a Yes

b $t = \sqrt{\frac{120 - d}{4.9}}$

c No solution provided.

d No solution provided.

e Yes

f 4 seconds